

1. Threat

- A **potential danger** that can exploit a weakness in a system and cause harm.
- **Source of harm.**

👉 Example:

- Hacker sending a phishing email.
- Flood damaging a data center.

👉 **In short:** A *threat* is **something bad that *could* happen.**

2. Vulnerability

- A **weakness** in a system, software, or process that can be exploited by a threat.
- Like an “open door” for attackers.

👉 Example:

- Using weak passwords.
- Unpatched software (Windows not updated).
- Misconfigured firewall.

👉 **In short:** A *vulnerability* is a **weak point.**

3. Risk

- The **likelihood** that a threat will exploit a vulnerability and the **impact** it will cause.
- Risk = **Threat × Vulnerability × Impact.**

👉 Example:

- Threat = Hacker trying to steal data.
- Vulnerability = Weak password (like "123456").
- Risk = High chance that the hacker will succeed and company data will be stolen.

👉 **In short:** *Risk* is **how much danger you face if the threat happens.**

Analogy (Easy Way to Remember)

- **Threat = Thief** (bad actor who can cause harm)
- **Vulnerability = Unlocked Door** (weakness the thief can use)
- **Risk = Chance of Getting Robbed** (thief exploiting the unlocked door and stealing valuables)